Machine Learning Group Project

Teams of 4–5 take 2-3 datasets from raw data to three evaluated models, and publish the work as a reproducible GitHub project.

Learning objectives

1. Apply the full ML workflow independently, from problem framing to evaluated model, in a group setting.
2. Produce three working models (one classification, one regression, one unsupervised) on a chosen dataset.
3. Organize and publish a reproducible project to GitHub.

1. Choose 2-3 datasets

Your team picks 2-3 datasets and use it for all three models. It must:

- have at least 1,000 rows and 8 features, with a mix of numeric and categorical columns
- contain, or let you build, a categorical target (classification) and a numeric target (regression)
- have meaningful features for grouping records (clustering or dimensionality reduction)
- be public, with a clear source and licence (Kaggle, UCI, data.gov, or similar)

Examples: student performance, heart disease, used-car listings, Airbnb listings, Spotify tracks, bank customers. You may bin a numeric column to create a class label, for example price into low / mid / high. Get the dataset approved before you start modelling.

2. Required work

Stage	What to deliver
Problem framing	Business or research question, why each model matters, target variables, success metrics, assumptions and limits.
Data understanding	Data dictionary, missing values, outliers, distributions, correlations, class balance. At least five meaningful plots, each with a written takeaway.
Preparation	Cleaning, encoding, scaling and feature engineering inside a pipeline. Train/test split made before any fitting, with a fixed random seed. No data leakage.
Classification	A baseline plus at least two algorithms. Cross-validation and hyperparameter tuning. Report accuracy, precision, recall, F1 and a confusion matrix; add ROC-AUC if the problem is binary.
Regression	A baseline plus at least two algorithms. Cross-validation and tuning. Report MAE, RMSE and R^2, plus a residual plot.
Unsupervised	Clustering (for example K-Means plus one other method), justification for the number of clusters (elbow, silhouette), and a profile of each cluster. PCA or t-SNE/UMAP for visualisation is encouraged.
Interpretation	Compare models against the baseline, show feature importance for one model, and analyse errors. State what you would do next and what the model should not be used for.

> **Keep it fair.** Never tune or choose models on the test set. Use cross-validation on the training data, then evaluate once on the held-out test set.

3. Team setup

Every member contributes to code, GitHub and the final presentation. Assign a lead for each area below, and rotate leads between milestones if you can.

- **Project coordinator:** keeps the schedule, runs meetings, merges pull requests.
- **Data lead:** data quality, EDA and the preprocessing pipeline.
- **Modelling leads:** one lead per model type. In a team of 4, one person covers two.
- **Repository and report lead:** README, folder structure, environment file, final write-up.

Work on feature branches and merge through pull requests with at least one reviewer. Each member should have visible commits and at least one reviewed pull request; the commit history is part of the assessment.

4. GitHub repository

Someone who has never seen your project should be able to clone it and reproduce your results by following the README.

```
project-name/
├── README.md            overview, results summary, how to run
├── requirements.txt     pinned versions
├── .gitignore
├── data/
│   ├── raw/             original data (or download script)
│   └── processed/
├── notebooks/           01_eda, 02_classification,
│                        03_regression, 04_clustering
├── src/                 reusable code: preprocessing, training, evaluation
├── models/              saved models (small files only)
├── reports/             figures and final report / slides
└── CONTRIBUTORS.md      who did what
```

Your README must include: the problem statement, dataset source, a results table for all three models, setup and run instructions, and team members with roles. Set random seeds, keep notebooks runnable from top to bottom, and never commit passwords or API keys. If the dataset is large, add a script to download it instead of committing it.

5. Timeline

Adjust the weeks to your course calendar.

Week	Milestone	Output
12	Proposal	Team, dataset, three problem statements, repo created
12	Data understanding	EDA notebook and data-quality notes
13	Preparation and baselines	Pipeline and baseline for each model
13	Modelling and tuning	Tuned classification, regression and clustering models
14	Interpretation and repository	Final README, report draft, clean repo
14	Presentation and submission	20-minute presentation with Q&A, repo link submitted

6. Submission

- Link to the public GitHub repository
- Presentation slides, and a live presentation where every member speaks

7. Grading (100 points)

Criteria	What earns full marks	Points
Problem framing and data understanding	Clear questions and metrics; insightful EDA	**15**
Data preparation	Leak-free pipeline, justified choices	**10**
Classification model	Baseline, comparison, tuning, correct metrics	**15**
Regression model	Baseline, comparison, tuning, correct metrics	**15**
Unsupervised model	Justified cluster count, meaningful cluster profiles	**15**
Interpretation and critical thinking	Honest limitations, error analysis, next steps	**10**
GitHub and reproducibility	Clean structure, working README, fresh-clone run succeeds	**10**
Report and presentation	Clear, organised, every member participates	**10**

Group marks are adjusted by individual contribution.

8. Ground rules

- You may use libraries (scikit-learn, pandas, and so on) and AI tools for help, but you must understand and be able to explain every line you submit. Any member can be asked about any part of the project.
- Cite your dataset and any code or ideas taken from elsewhere.
- Report your results honestly, including weak ones. A well-explained modest result scores higher than an unexplained high score.
- Raise team problems with your instructor early, not at the deadline.